

AERO40007 – Mechanics

Imperial College of Science, Technology & Medicine

Department of Aeronautics

1. Figure 1 shows three identical rough cylinders supported on a rough horizontal plane. Draw the free body diagrams for the three cylinders together and individually. Hence determine the smallest value of μ for equilibrium.

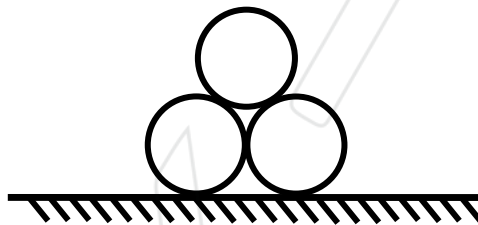
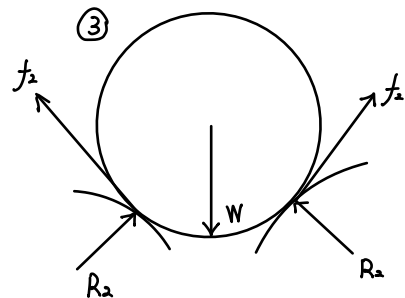
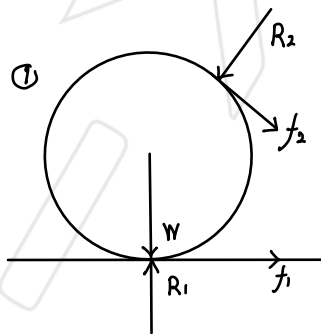
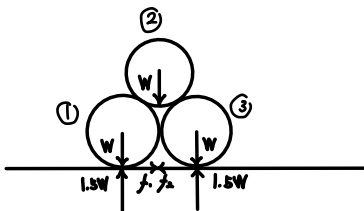


Figure 1.

ANS. The minimum value required is $\mu = 0.268$ 

$$\begin{aligned}\sum M &= 0 \\ &= f_1 r - f_2 r = 0\end{aligned}$$

$$\therefore f_1 = f_2 = f$$

$$\therefore \mu R_1 = \mu R_2 \quad \therefore R_1 = R_2 = R$$

$$\begin{aligned}\sum F &= 0 \\ \sum F_x &= 0 = f + f \cos 30^\circ - R \sin 30^\circ \\ &= \left(1 + \frac{\sqrt{3}}{2}\right) f - \frac{1}{2} R\end{aligned}$$

$$\therefore f = \mu R \quad \therefore \mu = \frac{1}{2} \left(1 + \frac{\sqrt{3}}{2}\right)^{-1} = 2 - \sqrt{3} = 0.26795 = 0.268$$

2. Figure 2 shows two identical blocks which are freely hinged along a common edge A. One of the blocks is supported at an accurate angle θ to the horizontal by a smooth vertical wall, while the other block rests on a rough horizontal plane. The wall support is above the level of A. If the coefficient of friction between block and plane is μ , show that slipping will be prevented if

$$\mu \geq \frac{1}{4} \left(\frac{1+2\alpha t - t^2}{2t + \alpha t^2 - \alpha} \right) \text{ where } t = \tan \frac{\theta}{2}$$

Derive in terms of μ , α and l , an expression for the longest distance between A and the line of action of the normal reaction between the block and plane.

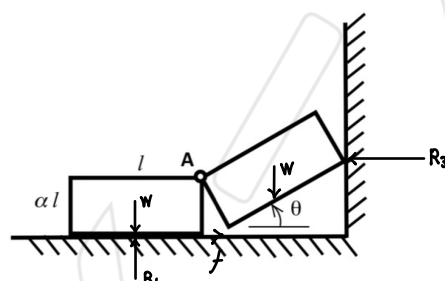
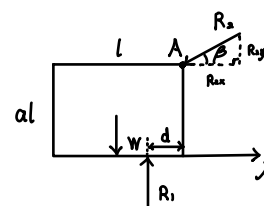
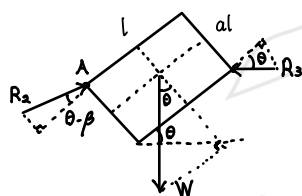


Figure 2.

$$\uparrow R_1 = W + W = 2W$$

$$\rightarrow R_3 = f = \mu R_1$$



$$\begin{aligned} \sum M_A &= (W \sin \theta) \left(\frac{1}{2} \alpha l \right) + (W \cos \theta) \left(\frac{1}{2} l \right) - (R_3 \sin \theta) l + (R_2 \cos \theta) \alpha l \\ &= W \left(\frac{1}{2} \sin \theta \alpha l + \frac{1}{2} \cos \theta l \right) + f (\cos \theta \alpha l - \sin \theta l) \\ &= \frac{1}{2} R_1 \left(\frac{1}{2} \sin \theta \alpha l + \frac{1}{2} \cos \theta l \right) + \mu R_1 (\cos \theta \alpha l - \sin \theta l) \\ &= \left(\frac{1}{4} \sin \theta \alpha l + \frac{1}{4} \cos \theta l \right) + \mu (\cos \theta \alpha l - \sin \theta l) = 0 \end{aligned}$$

$$\therefore \mu (\sin \theta - \cos \theta \alpha) = \left(\frac{1}{4} \sin \theta \alpha l + \frac{1}{4} \cos \theta l \right)$$

$$\therefore \mu (\tan \theta - \alpha) = \left(\frac{1}{4} \tan \theta \alpha l + \frac{1}{4} l \right)$$

$$\tan \theta = \tan \left(\frac{\theta}{2} + \frac{\theta}{2} \right) = \frac{\tan \frac{\theta}{2} + \tan \frac{\theta}{2}}{1 - \tan^2 \left(\frac{\theta}{2} \right)} = \frac{t + t}{1 - t^2} = \frac{2t}{1 - t^2}$$

$$\therefore \mu l \left(\frac{2t}{1 - t^2} - \alpha \right) = \frac{1}{4} l \left(\frac{2\alpha t}{1 - t^2} + 1 \right)$$

$$\therefore \mu \left(\frac{2t - \alpha + \alpha t^2}{1 - t^2} \right) = \frac{1}{4} \left(\frac{2\alpha t + 1 - t^2}{1 - t^2} \right)$$

$$\therefore \mu = \frac{1}{4} \left(\frac{2\alpha t + 1 - t^2}{2t - \alpha + \alpha t^2} \right)$$

$$\begin{aligned} \sum M_A &= f \times \alpha l + W \times \left(\frac{1}{2} l \right) - R_1 \times d \\ &= f \alpha l + \frac{1}{2} W l - R_1 d = 0 \end{aligned}$$

$$\therefore R_1 d = \frac{1}{2} W l + f \alpha l$$

$$\therefore R_1 d = \frac{1}{4} R_1 l + \mu R_1 \alpha l$$

$$\therefore d = \frac{1}{4} l + \mu \alpha l$$